NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-22885/89A(USAF)
18 February 1977
SUPERSEDING
MIL-S-22885/89(USAF)
26 January 1976

PERFORMANCE SPECIFICATION SHEET

SWITCH, PUSHBUTTON, ILLUMINATED,
SINGLE LAMP, 7 AMPERES, MOMENTARY ACTION

This specification is approved for use by the Engineering and Technical Support Division, Department of the Air Force (AFLC/LOIE), and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of specification MIL-S-22885.

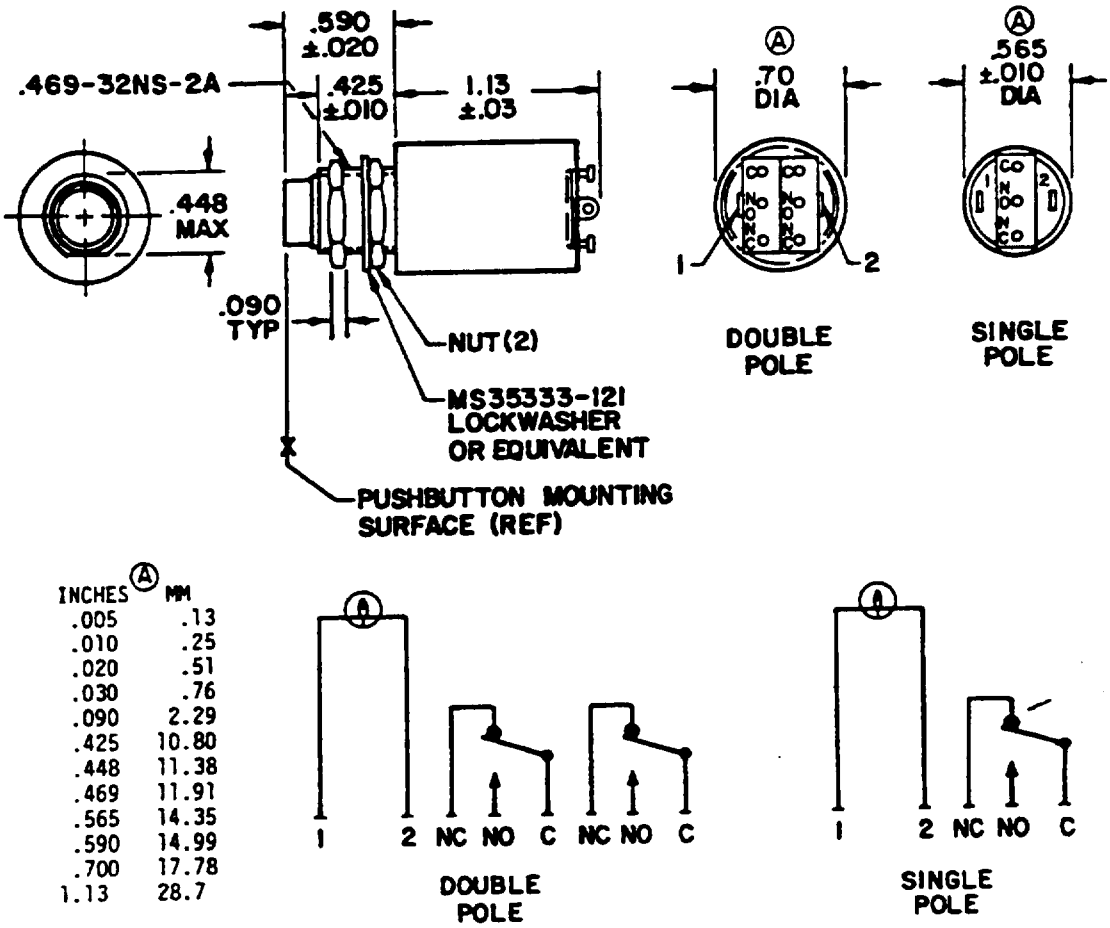
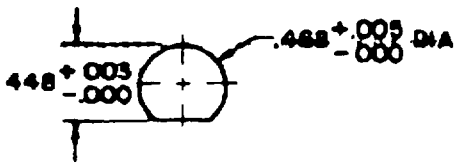


FIGURE 1. Basic switch body less lens.

FSC 5930

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"D" PATTERN HOLE SHOWN FOR BODY KEYING.
468 DIA. ROUND HOLE MAY ALSO BE USED

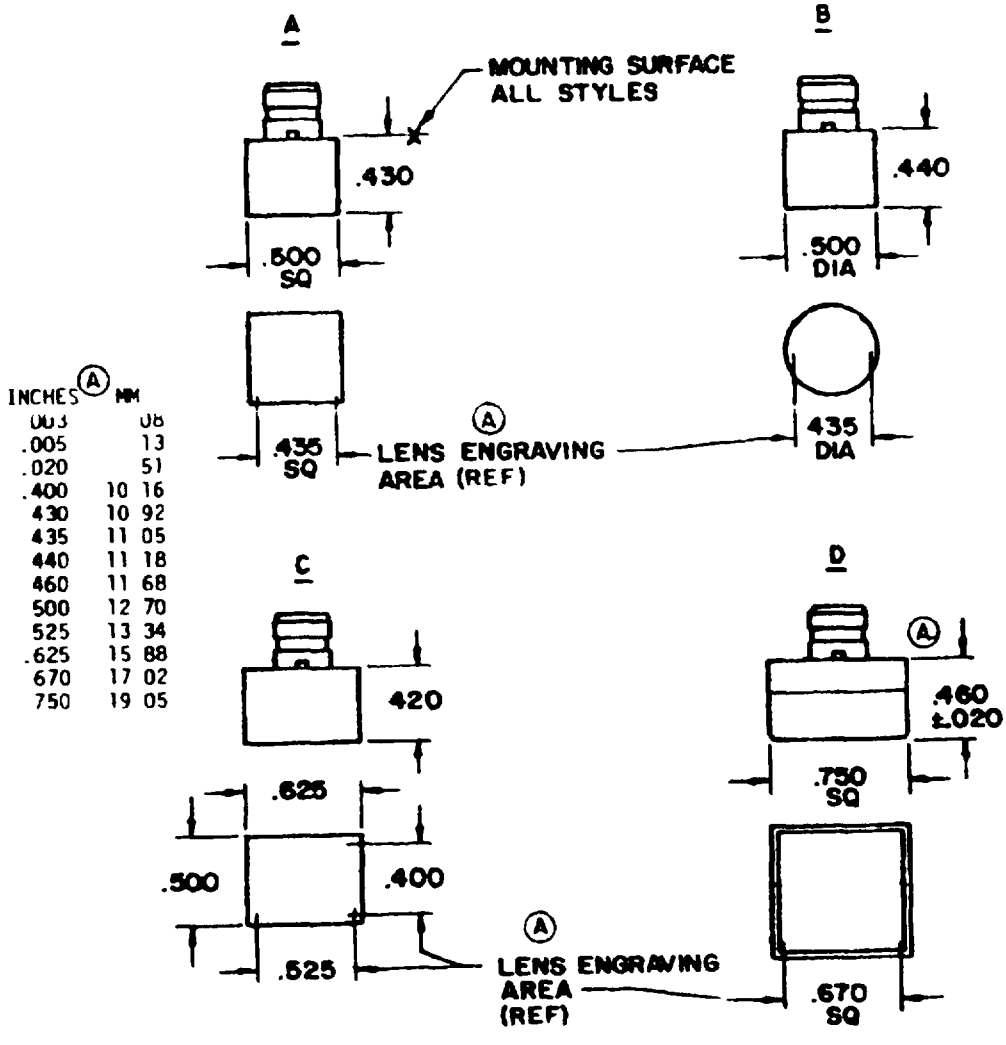
Recommended panel mounting hole dimensions.

INCHES	MM
.003	.08
.005	.13
448	11.38
.468	11.89

NOTES:
1 Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25 4 mm.
3 Unless otherwise specified, tolerances are ±.010 (25 mm) for two place decimals and ±.005 (.13 mm) for three place decimals.
4. Unit mounts in panels from .010 (.25 mm) to .20 (5.1 mm) thick.
5. Basic switches shall be MS24547-1 or MS24547-2 as listed on QPL-8805.
6 Terminals shall be permanently identified as shown in schematic

FIGURE 1 Basic switch body less lens - Continued.

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INCHES (A)	MM
.003	.08
.005	.13
.020	.51
.400	10.16
.430	10.92
.435	11.05
.440	11.18
.460	11.68
.500	12.70
.525	13.34
.625	15.88
.670	17.02
.750	19.05

NOTES
1 Lens shows color in both illuminated and nonilluminated conditions
2 Lens style D illuminates in forward direction only

FIGURE 2. Lens styles

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REQUIREMENTS

Dimensions and configuration See figures 1 and 2

Complete switch shall consist of:
1 - Lens, switch actuating, in accordance with figure 2.
1 - Lamp, T-1-3/4 midget flange based incandescent-not included, order separately
1 - Switch body in accordance with figure 1.

Enclosure design 1 (unsealed).

Temperature characteristic 1 (-55°C to +85°C)

Vibration grade. 3 (10-2,000 Hz)

Ⓐ Shock:
Lens style D: 40 G, 6 milliseconds, half-sine pulse.

Luminance and color values: See table III

Lens transmittance T (translucent).

Weight (less lamp):
1 pole: .04 pound maximum.
2 pole. .05 pound maximum

Operating characteristics.
Actuation force. 30 ±10 ounces
Actuation travel. .120 inches, maximum

Lens extraction force. 4 $^{+4}_{-2}$ pounds

Ⓐ Lens orientation. The lens shall withstand rotational torque up to 2 inch-pounds in either direction without rotating more than 8°

Mounting torque: 5 ±.5 inch-pounds

Electrical ratings: See tables I and II

TABLE I. Electrical ratings - MS24547-1 basic switch (silver contacts)

Load	Sea level		50,000 feet 28 Vdc (amperes)
	28 Vdc (amperes)	115 Vac, 60 Hz (amperes)	
Resistive	7	7	4
Inductive	4	7	2.5
Lamp	2.5	2	2.5

TABLE II Electrical ratings - MS24547-2 basic switch (gold contacts).

Load	Sea level	50,000 feet 28 Vdc (amperes)
	28 Vdc (amperes)	
Resistive	1	1
Inductive	5	.5

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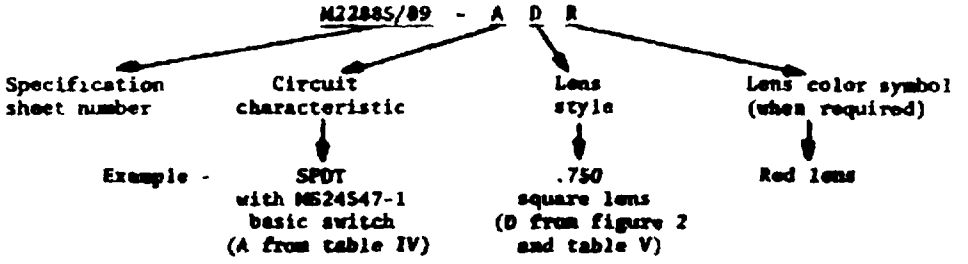
TABLE III. Luminance and color.

Color code	Luminance 1/ (footlamberts minimum average)	Chromaticity limits @ 2,360° Kelvin	
		X	Y
Green	36	.250 ±.030	.650 ± 030
Red	50	.685 ±.030	.310 ±.030
White	250	.525 ±.030	.413 ± 030
Yellow	250	.570 ±.030	.425 ± 030
Blue	20	.130 ±.030	345 ± 030

1/ When illuminated by one MS25237-327 lamp of 0.34 ±.02 mean spherical candle power.

Readings are for all lens styles.

Part number: Part numbers shall be assigned as illustrated below:



Part number M22885/89-ADR identifies a SPDT switch with silver contacts and a .750 square red lens.

TABLE IV. Circuit characteristics.

Code	Basic switch	Circuit
A	MS24547-1	SPDT
B	MS24547-1	DPDT
C	MS24547-2	SPDT
D	MS24547-2	DPDT
E 1/	None	---

1/ Use code E to specify lens portion of switch only.

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TABLE V. Lens style

Code	Lens style (see figure 2)
A	A
B	B
C	C
D	D
Z 1/	None

1/ Use code Z to specify switch body only (less lens)

Qualification inspection All applicants for qualification approval shall demonstrate that each of their items conforms to all the requirements specified in the applicable documents singularly and in combination with all other previously qualified items, regardless of manufacturer

Group submission

(a) When the basic sensitive switches are qualified to MIL-S-8805, switches shall be tested in accordance with table VI.
(b) When the basic sensitive switches are not qualified to MIL-S-8805, the switch type for which qualification is sought, shall be submitted to the complete tests in accordance with table X of MIL-S-22885.

TABLE VI Qualification inspection (group submission).

Part number	Inspection table of MIL-S-22885		Extent of approval
	Group	Number of samples	
M22885/89 -BDW	I II III VI VIII	(All sample units) 4(from group I) 1/ 2(from group I) 2(from group I) 2/ 1 each color 3/	All
M22885/89 -CCW	I II VI VIII	(All sample units) 4(from group I) 1/ 2(from group I) 2/ 1 each color 4/	

1/ Insulation resistance Following step 6 of the final cycle of the moisture resistance test, the insulation resistance measurement shall not be less than 5 megohms
2/ Sea level, resistive dc load only
3/ One style D lens of each color shall be tested in any switch from group I
4/ One style C lens of each color shall be tested in any switch from group I

Custodian
Air Force - 85

Preparing activity
Air Force - 85

Review activity
Air Force - 90

(Project 5930-F472)

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